POTPRESS · TRACEABILITY MATRIX

PotPress traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potpress		
Source	docs/03b-traceability.md @ 75fe8c1		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.4	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-23	Amish Chadha	Generated by .kit/trace.py
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PotPress traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 13 user requirements, 13 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Reference filter	PPR-CAL-001, PPR-BLD-001	Model check; shrinkage test bar on the partner clay (ASTM C326 method); later measurement of fired pots	Met on paper: 12.30 L brim, 9.91 L working fired; molds 1.087 times the fired size; shrinkage still assumed
UR-002, UR-012	PR-002	Even wall	PPR-CAL-001, PPR-BLD-001, PPR-STM-001	Tolerance stack and guide design; later sectioned pots	At risk**: ±0.44 mm with hand-finished molds; template print error of 0.2 to 0.5 % not yet in the budget (STEM E2)
UR-001, UR-007	PR-003	Pressing force and frame strength	PPR-CAL-001, PPR-BLD-001, PPR-STM-001	Beam and pin calculation; later proof load	Met on paper: beams 190 MPa, pin 272 MPa, M16 bolts 117 MPa in shear, male flange 60 MPa (cast yield about 80); 0.66 mm at 10 t
UR-003	PR-004	Throughput	PPR-CAL-001, PPR-BLD-001, PPR-STM-001	Cycle breakdown; later timed trial	Met on paper, thin margin: 5.2 min, 69 pots per 6 h; handle force about 148 N at the 5.4 t limit with 85 % efficiency (STEM S8; EN 1005-3 assessment to do)
UR-003	PR-005	Mold opening	PPR-CAL-001, PPR-BLD-001	Kinematic layout	Met on paper: 310 mm opening against 291 mm needed (49 mm clear); 110 of 150 mm stroke
UR-008, UR-012	PR-006	Flow-rate QC	PPR-CAL-001, PPR-BLD-001	Gauge calibration by volume; later repeat tests	Met on paper; repeatability to verify at TRL 4
UR-005	PR-007	Molds made locally	PPR-CAL-001	Pattern split and finishing review with a foundry	At risk**: 16 segments of 243 mm fit a true 250 x 250 mm bed only (STEM E1); hand finishing unproven
UR-004, UR-006	PR-008	Frame built locally	PPR-CAL-001, PPR-BLD-001, PPR-STM-001	Part list and mass estimate	Met on paper: heaviest parts 39.6 kg (female mold with plate, platen); lifting set BOM item 22
UR-007	PR-009	Safe operation	PPR-CAL-001, PPR-BLD-001	Hazard review and guard layout in the model; ISO 13857 desk check signed by a competent person before any	Met on paper: 23 of 23 openings; desk check not yet signed (hold point);

User requirement	Product requirement	Title	Evidence	Verification	Status
				force above hand pressure (decided by Amish, 2026-10-02)	Table 7 and floor gap not checked (STEM E3)
UR-009	PR-010	Affordable	PPR-CAL-001, PPR-BLD-001	Priced BOM (`bom/bom.csv`)	Over the value-engineering target by $219** ($1,279)
UR-010	PR-011	Footprint	PPR-CAL-001, PPR-BLD-001, BOM	Model check	Met on paper: 940 x 700 x 1,828 mm guarded; at the 0.7 m depth limit with no margin
UR-011	PR-012	Product-safe materials	PPR-CAL-001, PPR-BLD-001, PPR-STM-001	Material list review; a chill sample of each melt checked by spark OES, or handheld XRF for lead, before pouring, recorded per heat	Met on paper; EN 601 limit values to confirm
UR-013	PR-013	Local spares	BOM	BOM review (inspection, TRL 3); regional availability confirmed with a partner factory (TRL 4)	Met on paper; regional availability to confirm (text decided by Amish, 2026-10-25, D8)

Gaps

None.